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(54) **NAD(P)-DEPENDENT RESPONSIVE ENZYMES, ELECTRODES AND SENSORS, AND METHODS FOR MAKING AND USING THE SAME**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,321,123 A 3/1982 Nakamura et al.
4,721,677 A 1/1988 Clark, Jr.
(Continued)

FOREIGN PATENT DOCUMENTS

CN 1735375 A 2/2006
CN 101849180 B 8/2017
(Continued)

OTHER PUBLICATIONS

Nakabayashi, Y., et al., "Evaluation of osmium(II) complexes as electron transfer mediators accessible for amperometric glucose sensors," Anal Sci 17(8):945-950, The Japan Society for Analytical Chemistry, Japan (Aug. 2001).

(Continued)

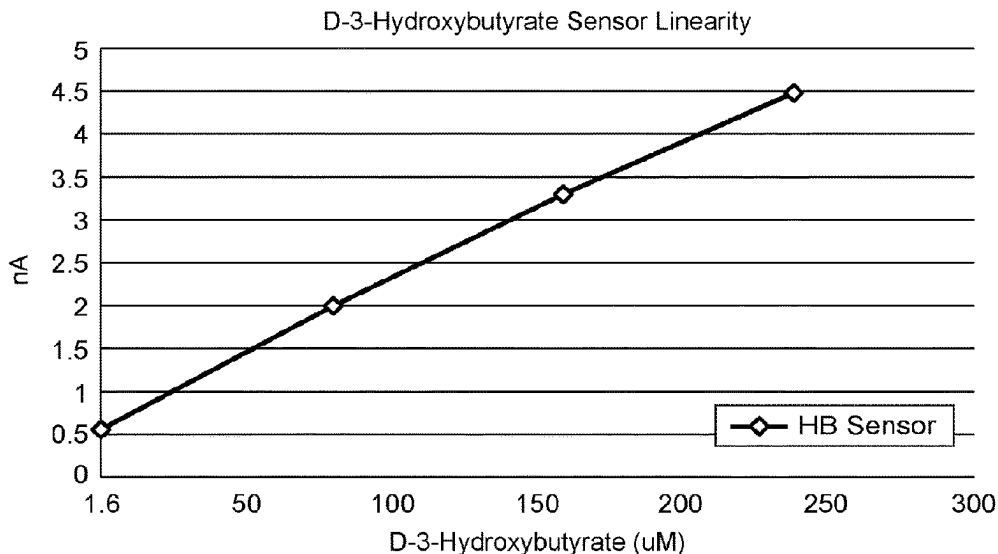
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(57) **ABSTRACT**

NADP-dependent oxidoreductase compositions, and electrodes, sensors and systems that include the same. Analyte sensors include an electrode having a sensing layer disposed thereon, the sensing layer comprising a polymer and an enzyme composition distributed therein. The enzyme composition includes nicotinamide adenine dinucleotide phosphate (NAD(P)⁺) or derivative thereof; an NAD(P)⁺-dependent dehydrogenase; an NAD(P)H oxidoreductase; and an electron transfer agent comprising a transition metal complex.

15 Claims, 3 Drawing Sheets



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References Cited

U.S. PATENT DOCUMENTS

- | | | | | | |
|---------------|---------|-----------------------------------|------------------|---------|---------------------------------------|
| 5,262,035 A | 11/1993 | Gregg et al. | 7,811,231 B2 | 10/2010 | Jin et al. |
| 5,264,105 A | 11/1993 | Gregg et al. | 7,822,557 B2 | 10/2010 | Chen et al. |
| 5,322,063 A | 6/1994 | Allen et al. | 8,106,780 B2 | 1/2012 | Goodnow et al. |
| 5,356,786 A | 10/1994 | Heller et al. | 8,268,143 B2 | 9/2012 | Liu et al. |
| 5,593,852 A | 1/1997 | Heller et al. | 8,435,682 B2 | 5/2013 | Heller |
| 5,696,314 A | 12/1997 | McCaffrey et al. | 8,444,834 B2 | 5/2013 | Liu et al. |
| 5,783,056 A | 7/1998 | Hamp et al. | 8,545,693 B2 | 10/2013 | McColl et al. |
| 5,792,621 A | 8/1998 | Verostko et al. | 8,761,857 B2 | 6/2014 | Feldman et al. |
| 5,820,551 A | 10/1998 | Hill et al. | 8,911,908 B2 | 12/2014 | Sakai et al. |
| 5,965,105 A | 10/1999 | Rayalu et al. | 9,290,839 B2 | 3/2016 | Wang et al. |
| 5,965,106 A | 10/1999 | Pomato et al. | 9,775,549 B2 | 10/2017 | Ouyang et al. |
| 6,071,391 A | 6/2000 | Gotoh et al. | 9,914,952 B2 | 3/2018 | Ouyang et al. |
| 6,121,009 A | 9/2000 | Heller et al. | 9,927,386 B2 | 3/2018 | Wang et al. |
| 6,134,461 A | 10/2000 | Say et al. | 9,983,161 B2 | 5/2018 | Feldman et al. |
| 6,175,752 B1 | 1/2001 | Say et al. | 10,022,076 B2 | 7/2018 | Hoss et al. |
| 6,241,863 B1 | 6/2001 | Monbouquette | 10,136,816 B2 | 11/2018 | Bernstein et al. |
| 6,284,478 B1 | 9/2001 | Heller et al. | 10,201,301 B2 | 2/2019 | Heller et al. |
| 6,377,894 B1 | 4/2002 | Deweese et al. | 10,702,193 B2 | 7/2020 | Simpson et al. |
| 6,503,381 B1 | 1/2003 | Gotoh et al. | 11,091,788 B2 | 8/2021 | Ouyang et al. |
| 6,514,460 B1 | 2/2003 | Fendrock | 2001/0003045 A1 | 6/2001 | Davis et al. |
| 6,514,718 B2 | 2/2003 | Heller et al. | 2003/0042137 A1 | 3/2003 | Mao et al. |
| 6,541,216 B1* | 4/2003 | Wilsey C12Q 1/005
435/26 | 2003/0050547 A1 | 3/2003 | Lebel et al. |
| 6,560,471 B1 | 5/2003 | Heller et al. | 2003/0068666 A1* | 4/2003 | Zweig C12Q 1/005
435/14 |
| 6,579,690 B1 | 6/2003 | Bonnecaze et al. | 2003/0100821 A1 | 5/2003 | Heller et al. |
| 6,600,997 B2 | 7/2003 | Deweese et al. | 2004/0061232 A1 | 4/2004 | Shah et al. |
| 6,605,200 B1 | 8/2003 | Mao et al. | 2005/0148832 A1 | 7/2005 | Reghabi et al. |
| 6,605,201 B1 | 8/2003 | Mao | 2005/0215871 A1 | 9/2005 | Feldman et al. |
| 6,654,625 B1 | 11/2003 | Say et al. | 2006/0004272 A1 | 1/2006 | Shah et al. |
| 6,676,816 B2 | 1/2004 | Mao et al. | 2006/0076236 A1 | 4/2006 | Shah et al. |
| 6,736,957 B1 | 5/2004 | Forrow et al. | 2006/0257996 A1 | 11/2006 | Simpson et al. |
| 6,746,582 B2 | 6/2004 | Heller et al. | 2007/0007132 A1 | 1/2007 | Mao et al. |
| 6,881,551 B2 | 4/2005 | Heller et al. | 2007/0027385 A1 | 2/2007 | Brister et al. |
| 6,932,894 B2 | 8/2005 | Mao et al. | 2007/0042377 A1 | 2/2007 | Gao et al. |
| 6,946,299 B2 | 9/2005 | Neel et al. | 2007/0095661 A1 | 5/2007 | Wang et al. |
| 7,090,756 B2 | 8/2006 | Mao et al. | 2007/0131547 A1* | 6/2007 | Nomoto H01M 8/16
204/403.01 |
| 7,299,082 B2 | 11/2007 | Feldman et al. | 2007/0213611 A1 | 9/2007 | Simpson et al. |
| 7,501,053 B2 | 3/2009 | Karinka et al. | 2007/0235331 A1* | 10/2007 | Simpson A61B 5/00
204/403.01 |
| 7,520,970 B2 | 4/2009 | Sato et al. | 2007/0270672 A1 | 11/2007 | Hayter |
| 7,563,588 B2 | 7/2009 | Gao et al. | 2008/0179187 A1 | 7/2008 | Ouyang et al. |
| 7,754,093 B2 | 7/2010 | Forrow et al. | 2008/0319295 A1 | 12/2008 | Bernstein et al. |
| | | | 2009/0166194 A1 | 7/2009 | Sato et al. |
| | | | 2009/0294306 A1 | 12/2009 | Feldman et al. |
| | | | 2010/0185070 A1 | 7/2010 | Brister et al. |
| | | | 2010/0198034 A1 | 8/2010 | Thomas et al. |
| | | | 2010/0213057 A1 | 8/2010 | Feldman et al. |
| | | | 2010/0230285 A1 | 9/2010 | Hoss et al. |
| | | | 2010/0267161 A1 | 10/2010 | Wu et al. |
| | | | 2010/0268043 A1* | 10/2010 | Yodfat A61M 5/1723
600/345 |
| | | | 2010/0324392 A1 | 12/2010 | Yee et al. |
| | | | 2010/0326842 A1 | 12/2010 | Mazza et al. |
| | | | 2011/0039165 A1 | 2/2011 | Sugiyama et al. |
| | | | 2011/0046467 A1 | 2/2011 | Simpson et al. |
| | | | 2011/0120865 A1 | 5/2011 | Bommakanti et al. |
| | | | 2011/0124993 A1 | 5/2011 | Bommakanti et al. |
| | | | 2011/0124994 A1 | 5/2011 | Bommakanti et al. |
| | | | 2011/0126188 A1 | 5/2011 | Bernstein et al. |
| | | | 2011/0213057 A1 | 9/2011 | Fenn et al. |
| | | | 2011/0213225 A1 | 9/2011 | Bernstein et al. |
| | | | 2011/0256024 A1 | 10/2011 | Cole et al. |
| | | | 2011/0257495 A1 | 10/2011 | Hoss et al. |
| | | | 2012/0132525 A1 | 5/2012 | Liu et al. |
| | | | 2012/0138484 A1 | 6/2012 | Bommakanti et al. |
| | | | 2012/0150005 A1 | 6/2012 | Hoss et al. |
| | | | 2012/0157801 A1 | 6/2012 | Hoss et al. |
| | | | 2012/0181189 A1 | 7/2012 | Merchant |
| | | | 2012/0186997 A1 | 7/2012 | Li et al. |
| | | | 2012/0245447 A1 | 9/2012 | Karan et al. |
| | | | 2012/0283537 A1 | 11/2012 | Petisce et al. |
| | | | 2012/0296186 A1 | 11/2012 | Ouyang et al. |
| | | | 2012/0323098 A1 | 12/2012 | Moein et al. |
| | | | 2013/0059212 A1 | 3/2013 | Kusumegi et al. |
| | | | 2013/0116524 A1 | 5/2013 | Cole et al. |
| | | | 2013/0131478 A1 | 5/2013 | Simpson et al. |
| | | | 2013/0211219 A1 | 8/2013 | Coppeta et al. |
| | | | 2013/0231542 A1 | 9/2013 | Simpson et al. |

(56)

References Cited**U.S. PATENT DOCUMENTS**

2013/0245412	A1	9/2013	Rong et al.
2013/0324820	A1	12/2013	Petillo et al.
2014/0054171	A1	2/2014	Feldman et al.
2014/0127728	A1	5/2014	Wilsey
2014/0176338	A1	6/2014	He et al.
2014/0262776	A1	9/2014	Martin et al.
2014/0262777	A1	9/2014	Zhao et al.
2015/0038814	A1	2/2015	Staib et al.
2015/0076004	A1	3/2015	Gerber et al.
2015/0207796	A1	7/2015	Love et al.
2016/0045147	A1	2/2016	Ouyang et al.
2016/0319232	A1	11/2016	Noritomi et al.
2016/0328991	A1	11/2016	Simpson et al.
2016/0345882	A1	12/2016	Wu et al.
2016/0354542	A1	12/2016	Ward et al.
2016/0355862	A1*	12/2016	Deng G01N 27/3271
2017/0156652	A1	6/2017	Qiang et al.
2017/0191958	A1	7/2017	Gerber et al.
2017/0202491	A1	7/2017	Heller et al.
2017/0315077	A1	11/2017	Rao et al.
2017/0319111	A1	11/2017	Simpson et al.
2018/0014766	A1	1/2018	Ouyang et al.
2018/0116604	A1	5/2018	Newberry
2018/0275088	A1	9/2018	Huff et al.
2019/0004005	A1	1/2019	Oja et al.
2019/0024130	A1	1/2019	Ouyang et al.
2019/0125230	A1	5/2019	Feldman
2019/0271658	A1	9/2019	Haneda et al.
2019/0274598	A1	9/2019	Scott et al.
2019/0320947	A1	10/2019	Chen et al.
2020/0069226	A1	3/2020	Hahn et al.
2020/0237275	A1	7/2020	Feldman et al.
2020/0237276	A1	7/2020	Oja et al.
2020/0237277	A1	7/2020	Ouyang et al.
2020/0241015	A1	7/2020	Ouyang et al.
2021/0137431	A1	5/2021	Oja et al.
2022/0056500	A1	2/2022	Ouyang et al.
2022/0168727	A1	6/2022	Baldwa
2022/0386910	A1	12/2022	Oja et al.
2022/0389475	A1	12/2022	Ouyang et al.
2022/0395202	A1	12/2022	Ouyang et al.
2022/0396820	A1	12/2022	Ouyang et al.
2023/0080107	A1	3/2023	Ouyang et al.
2023/0118818	A1	4/2023	Feldman et al.
2023/0119512	A1	4/2023	Feldman et al.
2023/0121101	A1	4/2023	Feldman et al.
2023/0121367	A1	4/2023	Feldman et al.
2023/0121769	A1	4/2023	Feldman et al.
2023/0122702	A1	4/2023	Feldman et al.
2023/0123384	A1	4/2023	Feldman et al.
2023/0128038	A1	4/2023	Feldman et al.
2024/0247297	A1	7/2024	Ouyang et al.

FOREIGN PATENT DOCUMENTS

CN	113164106	A	7/2021
EP	0990706	A1	4/2000
EP	3394252	A1	10/2018
EP	3414325	A1	12/2018
EP	3597765	A1	1/2020
EP	4084689	A1	11/2022
GB	2067764	A	7/1981
JP	2001506742	A	5/2001
JP	2007510155	A	4/2007
JP	2007290504	A	11/2007
JP	2010517054	A	5/2010
JP	2010530790	A	9/2010
JP	2011136186	A	7/2011
JP	2013504053	A	2/2013
JP	2014503080	A	2/2014
JP	2014160024	A	9/2014
JP	2015198960	A	11/2015
JP	2018029983	A	3/2018
JP	2022172249	A	11/2022
WO	WO-03056319	A2	7/2003

WO	WO-2005040404	A1	5/2005
WO	WO-2009105337	A2	8/2009
WO	WO-2010030912	A1	3/2010
WO	WO-2011030093	A1	3/2011
WO	WO-2012100130	A1	7/2012
WO	WO-2015150263	A1	10/2015
WO	WO-2015195352	A1	12/2015
WO	WO-2016025064	A1	2/2016
WO	WO-2016174456	A1	11/2016
WO	WO-2016174458	A1	11/2016
WO	WO-2017151952	A1	9/2017
WO	WO-2018067235	A1	4/2018
WO	WO-2019006413	A1	1/2019

OTHER PUBLICATIONS

Notice of Allowance mailed Apr. 13, 2021 in U.S. Appl. No. 16/081,162, 7 pages.

Supplementary European Search Report in EP Application No. 17760843, Feb. 26, 2020, 2 pages.

International Search Report and Written Opinion of PCT/US2017/020495, mailed Jul. 10, 2017, 23 pages.

Nikitina, O., et al., "Bi-enzyme biosensor based on NAD⁺- and glutathione-dependent recombinant formaldehyde dehydrogenase and diaphorase for formaldehyde assay," *Sensors and Actuators* 125(1): 1-9, (2007).

Office Action mailed Jul. 22, 2020 in U.S. Appl. No. 16/081,162, 10 pages.

Office Action mailed Dec. 30, 2020 in U.S. Appl. No. 16/081,162, 12 pages.

Non-Final Office Action mailed Mar. 25, 2024, in U.S. Appl. No. 17/403,258, 9 pages.

Non-Final Office Action mailed Jan. 24, 2024, in U.S. Appl. No. 17/818,143, 14 pages.

Non-Final Office Action mailed Mar. 12, 2024, in U.S. Appl. No. 17/818,770, 14 pages.

Extended European Search Report for European Application No. EP17760843, mailed on Mar. 5, 2020, 8 pages.

Final Office Action mailed Jul. 30, 2024, in U.S. Appl. No. 17/818,143, 11 pages.

Final Office Action mailed Jul. 30, 2024, in U.S. Appl. No. 17/818,770, 13 pages.

Final Office Action mailed Nov. 13, 2024, in U.S. Appl. No. 17/403,258, 11 pages.

Final Office Action mailed Oct. 25, 2024, in U.S. Appl. No. 17/818,912, 11 pages.

Non-Final Office Action mailed Apr. 26, 2024, in U.S. Appl. No. 17/818,912, 13 pages.

Non-Final Office Action mailed May 7, 2024, in U.S. Appl. No. 17/819,151, 13 pages.

Non-Final Office Action mailed Nov. 5, 2024, in U.S. Appl. No. 17/818,770, 14 pages.

Non-Final Office Action mailed Nov. 14, 2024, in U.S. Appl. No. 17/818,143, 12 pages.

Non-Final Office Action mailed Sep. 16, 2024, in U.S. Appl. No. 18/597,704, 9 pages.

Notice of Allowance mailed Oct. 22, 2024, in U.S. Appl. No. 17/819,151, 7 pages.

Burmeister, J.J., and Gerhardt, G.A., "Self-referencing Ceramic-based Multisite Microelectrodes for the Detection and Elimination of Interferences from the Measurement of L-glutamate and Other Analytes," *Analytical Chemistry*, 73(5): 1037-1042, American Chemical Society, United States (Mar. 2001).

Cardosi, M., and Liu, Z., "Amperometric Glucose Sensors for Whole Blood Measurement Based on Dehydrogenase Enzymes," Chapter 13 in *Dehydrogenases*, pp. 319-354, Canuto, R. A., ed., InTechOpen, Croatia, 36 pages (Nov. 2012).

D'Allegro, J., "Soon Your Car Will Know When You Are Having a Heart Attack—And Know How to React," *CNBC*, Accessed at <https://www.cnn.com/2017/11/17/cars-will-know-when-youre-having-a-heart-attack-and-how-to-react.html>, Accessed on Oct. 9, 2023, (Nov. 17, 2017), 7 Pages.

English language translation of JP 2014-160024 A, 22 pages.

(56)

References Cited**OTHER PUBLICATIONS**

Final Office Action for U.S. Appl. No. 16/582,583, mailed on Apr. 19, 2023, 18 pages.
 Final Office Action for U.S. Appl. No. 16/582,583, mailed on May 16, 2024, 21 pages.
 Final Office Action for U.S. Appl. No. 16/774,835, mailed on Apr. 18, 2022, 24 pages.
 Final Office Action for U.S. Appl. No. 16/774,835, mailed on Dec. 24, 2024, 25 pages.
 Final Office Action for U.S. Appl. No. 16/774,909, mailed on Apr. 9, 2024, 13 pages.
 Final Office Action for U.S. Appl. No. 17/138,477, mailed on Mar. 17, 2023, 12 pages.
 Final Office Action for U.S. Appl. No. 17/138,477, mailed on Apr. 3, 2024, 12 pages.
 Final Office Action for U.S. Appl. No. 18/068,019, mailed on Oct. 17, 2024, 17 pages.
 Final Office Action for U.S. Appl. No. 18/068,072, mailed on Oct. 18, 2024, 13 pages.
 Final Office Action for U.S. Appl. No. 18/068,077, mailed on Jan. 28, 2025, 26 pages.
 Final Office Action for U.S. Appl. No. 18/068,704, mailed on Jan. 24, 2025, 16 pages.
 Final Office Action for U.S. Appl. No. 18/068,714, mailed on Oct. 18, 2024, 15 pages.
 Final Office Action for U.S. Appl. No. 18/068,860, mailed on Oct. 18, 2024, 15 pages.
 Final Office Action for U.S. Appl. No. 16/774,835, mailed on Sep. 12, 2023, 16 pages.
 Final Office Action for U.S. Appl. No. 16/774,909, mailed on Dec. 21, 2022, 13 pages.
 Guiseppi-Elie, A., et al., "Design of a subcutaneous implantable biochip for monitoring of glucose and lactate," *IEEE Sensors Journal* 5(3):345-355, Institute of Electrical and Electronics Engineers, United States (Jun. 2005).
 International Search Report and Written Opinion for International Application No. PCT/US2019/052942, dated on Nov. 28, 2019, 17 pages.
 International Search Report and Written Opinion for International Application No. PCT/US2020/015365, mailed May 28, 2020, 14 pages.
 International Search Report and Written Opinion for International Application No. PCT/US2020/015400, mailed on Apr. 9, 2020, 14 pages.
 International Search Report and Written Opinion for International Application No. PCT/US2020/015321, mailed on Apr. 9, 2020, 14 pages.
 Monteiro, T., et al., "Construction of Effective Disposable Biosensors for Point of Care Testing of Nitrite," *Talanta* 142:246-251, Elsevier, Netherlands (Sep. 2015).
 Mueller, S., et al., "The GOX/CAT system: A Novel Enzymatic Method to Independently Control Hydrogen Peroxide and Hypoxia in Cell Culture" *Advances in Medical Sciences* 54(2):121-135, Elsevier, Netherlands (Dec. 2009).
 Non-Final Office Action for U.S. Appl. No. 16/582,583, mailed on Oct. 3, 2023, 22 pages.
 Non-Final Office Action for U.S. Appl. No. 16/582,583, mailed on Sep. 1, 2022, 20 pages.
 Non-Final Office Action for U.S. Appl. No. 16/774,835, mailed on Apr. 15, 2024, 16 pages.
 Non-Final Office Action for U.S. Appl. No. 16/774,835, mailed on Sep. 17, 2021, 21 pages.

Non-Final Office Action for U.S. Appl. No. 16/774,841, mailed on Jun. 24, 2022, 10 pages.
 Non-Final Office Action for U.S. Appl. No. 16/774,841, mailed on Nov. 16, 2022, 10 pages.
 Non-Final Office Action for U.S. Appl. No. 16/774,909, mailed on Aug. 8, 2023, 14 pages.
 Non-Final Office Action for U.S. Appl. No. 16/774,909, mailed on Jun. 2, 2022, 13 pages.
 Non-Final Office Action for U.S. Appl. No. 17/138,477, mailed on Sep. 12, 2022, 10 pages.
 Non-Final Office Action for U.S. Appl. No. 17/138,477, mailed on Aug. 21, 2023, 12 pages.
 Non-Final Office Action for U.S. Appl. No. 17/151,274, mailed on Nov. 25, 2022, 13 pages.
 Non-Final Office Action for U.S. Appl. No. 17/151,274, mailed on Sep. 7, 2023, 14 pages.
 Non-Final Office Action for U.S. Appl. No. 17/819,099, mailed on Mar. 5, 2024, 14 pages.
 Non-Final Office Action for U.S. Appl. No. 17/819,099, mailed on Sep. 22, 2023, 24 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,019, mailed on Feb. 29, 2024, 12 Pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,072, mailed on Feb. 15, 2024, 13 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,077, mailed on Jul. 30, 2024, 14 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,704, mailed on Jun. 18, 2024, 14 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,714, mailed on Mar. 5, 2024, 14 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,807, mailed on Jan. 29, 2025, 25 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,834, mailed on Jan. 31, 2024, 10 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,834, mailed on Oct. 18, 2024, 15 pages.
 Non-Final Office Action for U.S. Appl. No. 18/068,860, mailed on Jul. 15, 2024, 13 pages.
 Non-Final Office Action for U.S. Appl. No. 16/774,835, mailed on Mar. 2, 2023, 14 pages.
 Notice of Allowance for U.S. Appl. No. 16/774,841, mailed on Aug. 16, 2023, 8 pages.
 Notice of Allowance for U.S. Appl. No. 16/774,841, mailed on Feb. 9, 2024, 3 pages.
 Notice of Allowance for U.S. Appl. No. 16/774,909, mailed on Sep. 29, 2024, 10 pages.
 Notice of Allowance for U.S. Appl. No. 17/151,274, mailed on Feb. 7, 2024, 8 pages.
 Notice of Allowance for U.S. Appl. No. 17/819,099, mailed on Jul. 9, 2024, 8 pages.
 Pickup, J.C., "Glucose Sensors: Present and Future," *International Textbook of Diabetes Mellitus*, Third Edition, DeFronzo, R.A., eds., 2:1686-1694, John Wiley & Sons Inc., United States (2004), 13 Pages.
 Pundir, C.S., et al., "Biosensing Methods for Determination of Creatinine: A Review," *Biosensors and Bioelectronics Journal* 126:707-724, Elsevier Advanced Technology, United Kingdom (Feb. 2019).
 Shi, G., et al., "The Study of Nafion/xanthine Oxidase/au Colloid Chemically Modified Biosensor and Its Application in the Determination of Hypoxanthine in Myocardial Cells in Vivo," *The Analyst* 127(3):396-400, Royal Society of Chemistry, United Kingdom (Mar. 2002).

* cited by examiner

FIG. 1

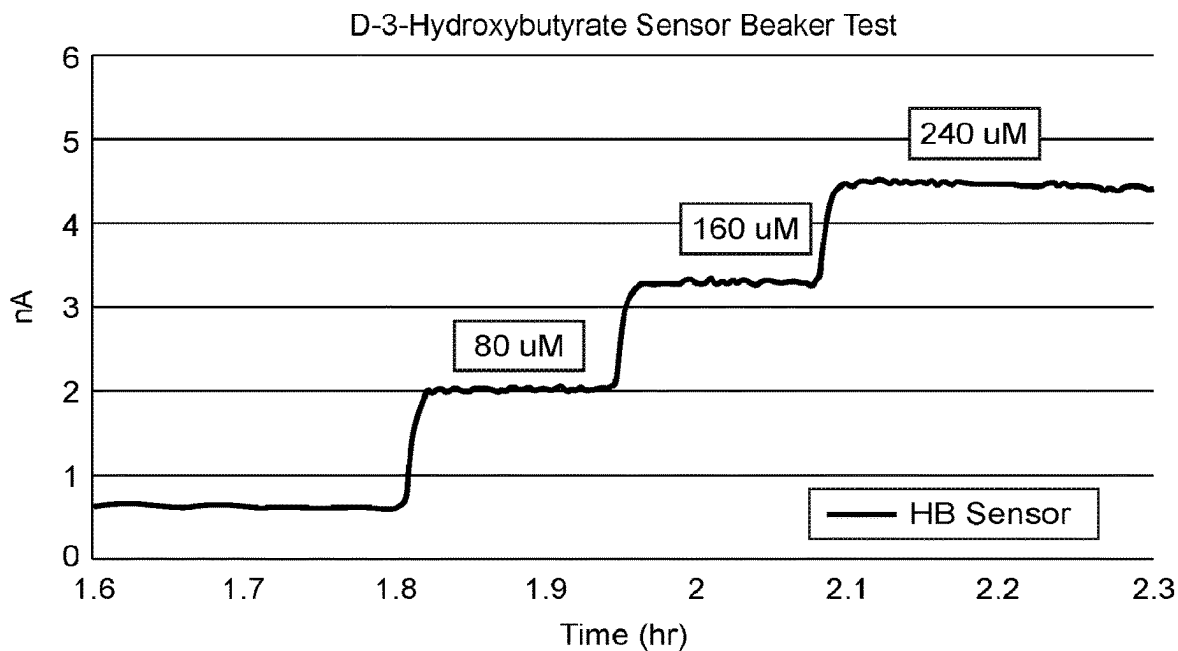


FIG. 2

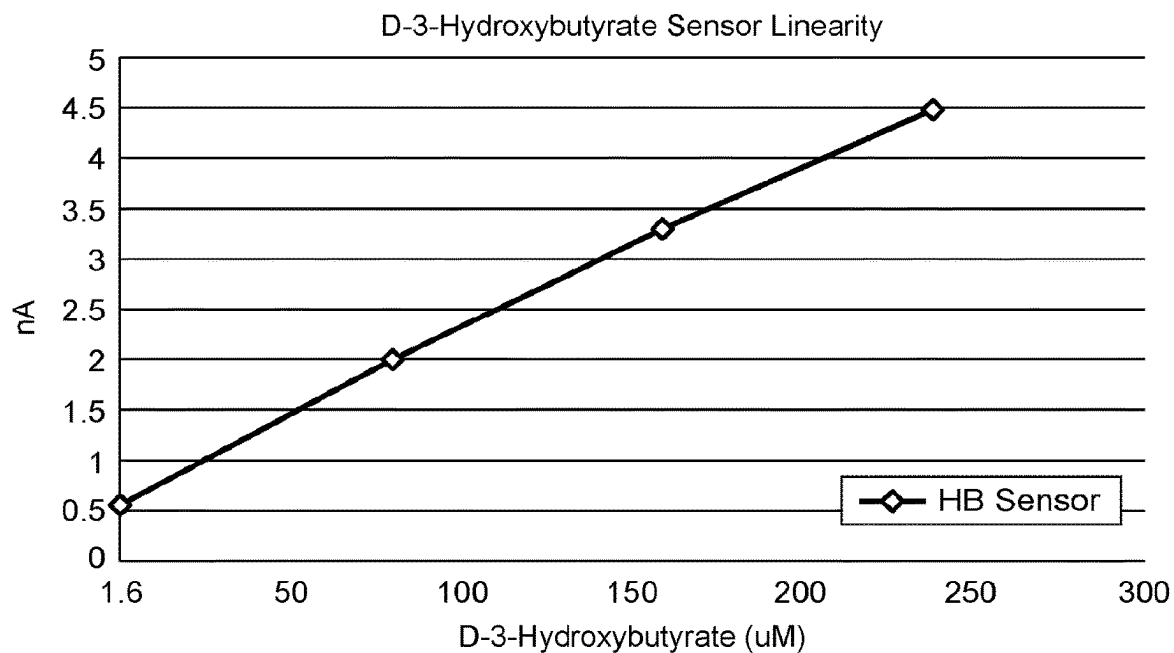


FIG. 3

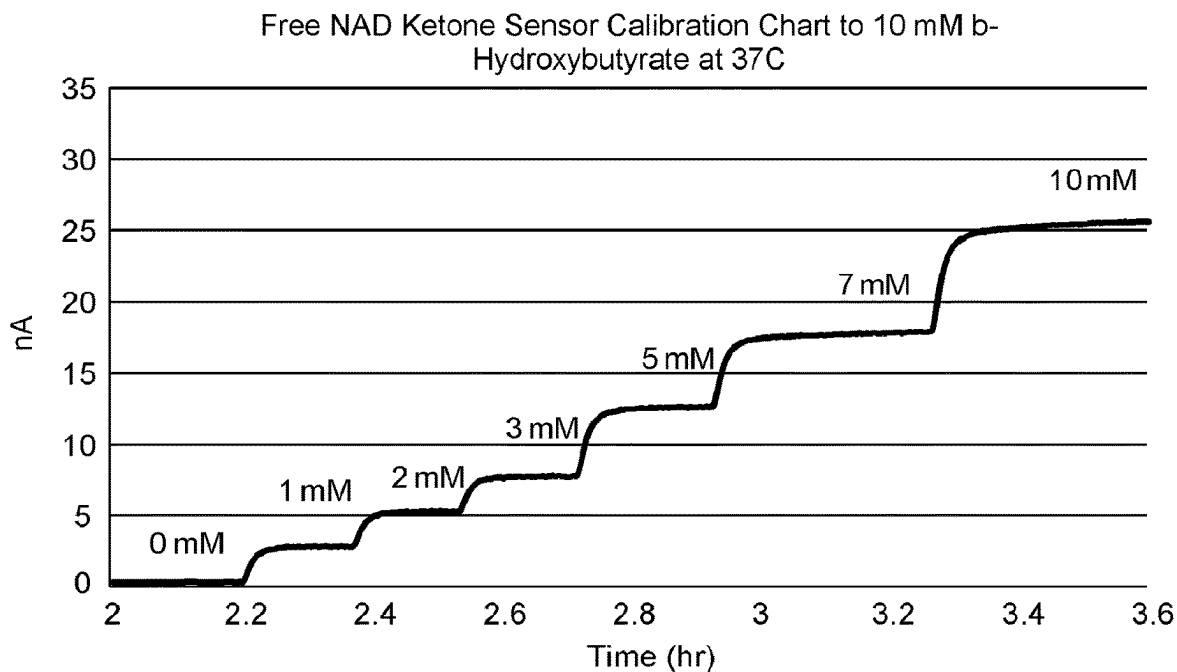


FIG. 4

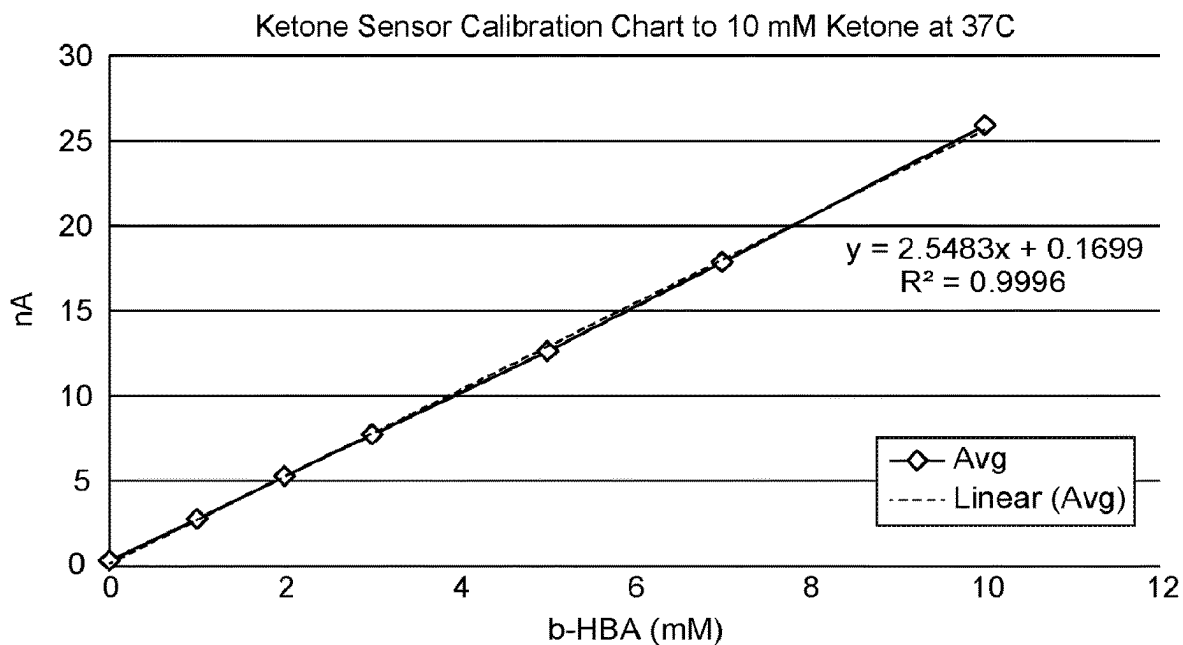


FIG. 5

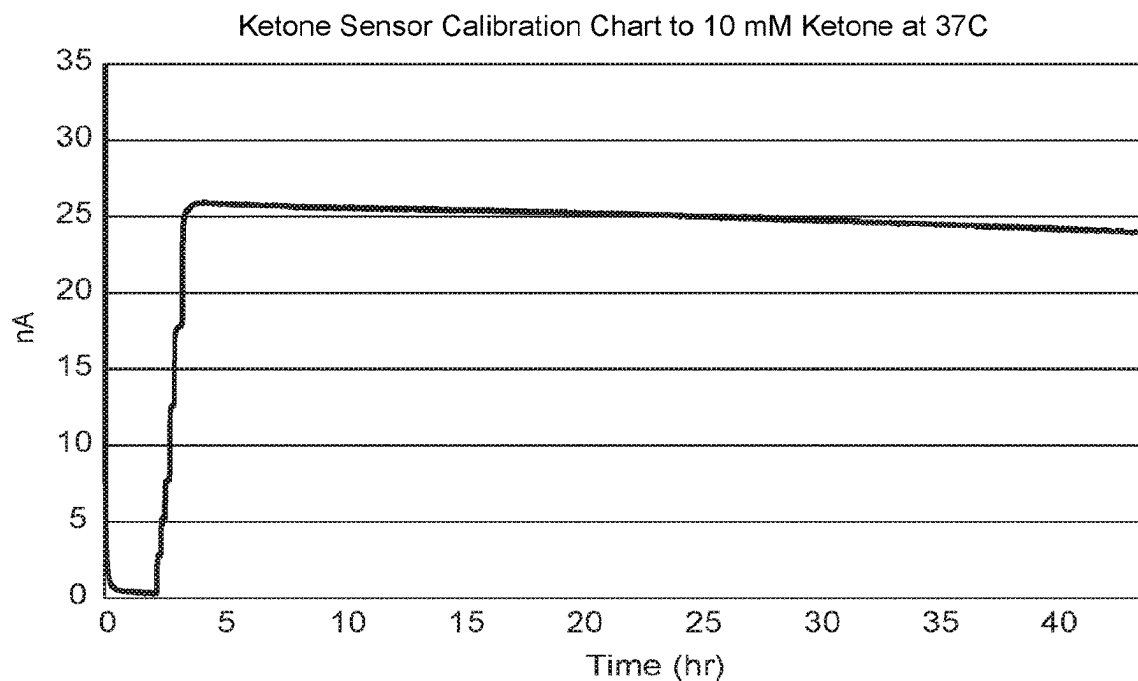
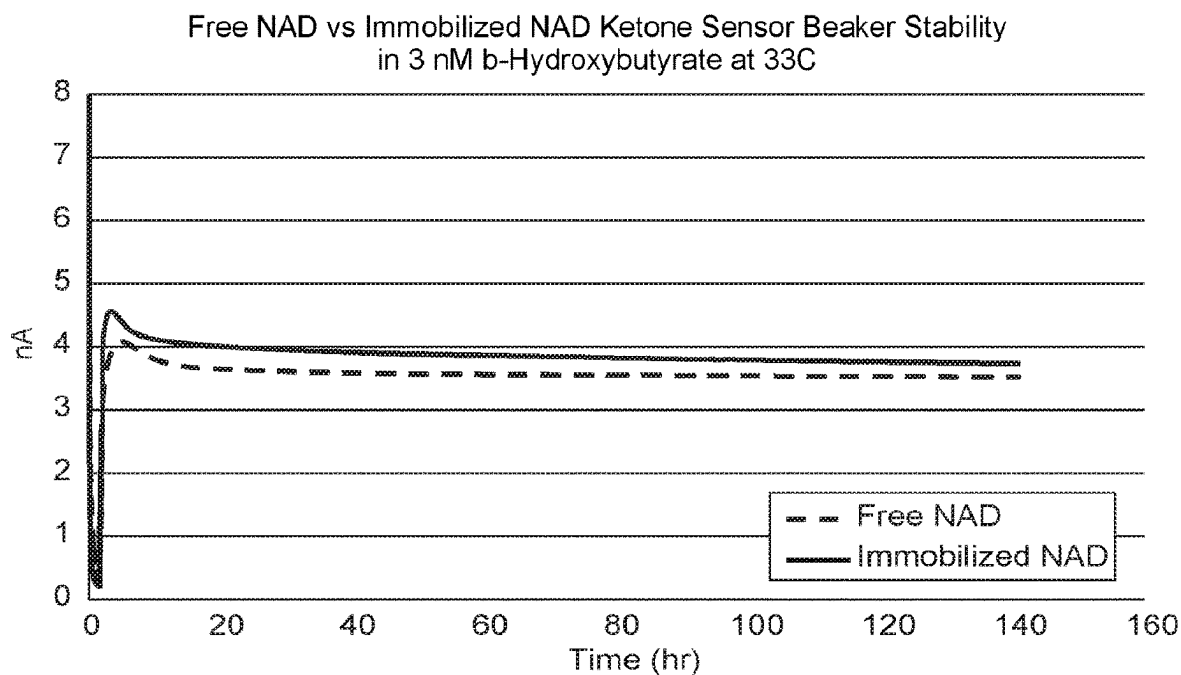


FIG. 6



NAD(P)-DEPENDENT RESPONSIVE ENZYMES, ELECTRODES AND SENSORS, AND METHODS FOR MAKING AND USING THE SAME

INTRODUCTION

The characterization of analytes in biological fluids has become an integral part of medical diagnoses and assessments of overall health and wellness of a patient. Regularly monitoring the concentration of particular analytes in body fluid of a subject is becoming increasingly important where the results may play a prominent role in the treatment protocol of a patient in a variety of disease conditions. A number of systems are available that analyze an analyte in a bodily fluid, such as blood, plasma, serum, interstitial fluid, urine, tears, and saliva. Such systems monitor the level of particular medically relevant analytes, such as, blood sugars, e.g., glucose, cholesterol, ketones, vitamins, proteins, and various metabolites. In response to this growing importance of analyte monitoring, a variety of analyte detection protocols and devices for laboratory, clinical and at-home use have been developed.

Nicotinamide adenine dinucleotide (NAD(P)+) is a coenzyme found in all living cells. There are many biological molecules that are oxidized by NAD(P)+-dependent dehydrogenases. For example, glucose can be oxidized by NAD-dependent glucose dehydrogenase, alcohol can be oxidized by NAD-dependent alcohol dehydrogenase, β -Hydroxybutyrate can be oxidized by NAD-dependent D-3-Hydroxybutyrate dehydrogenase, etc. Development of improved sensors for measuring analytes that are oxidized by a variety of different enzymes having a high degree of stability and sensitivity is desirable.

SUMMARY

Embodiments of the present disclosure relate to enzyme compositions, electrodes, sensors, methods for fabricating an analyte sensor as well as methods for monitoring an analyte concentration in a subject with the subject sensors. Disclosed herein are enzyme compositions that include one or more of nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, an NAD(P)+-dependent dehydrogenase, an NAD(P)H oxidoreductase and an electron transfer agent having a transition metal complex. In certain embodiments, enzyme compositions include NAD(P)+ or a derivative thereof and an electron transfer agent having a transition metal complex. The subject enzyme compositions may also include a polymer, such as a heterocycle-containing polymer, and a crosslinker for immobilizing the enzyme composition onto a surface (e.g., on the surface of an electrode). Some or all of these components may be unbound or unconnected, or two or more of these components may be bound or connected together. For example, one or more of the nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, NAD(P)+-dependent dehydrogenase, NAD(P)H oxidoreductase and redox mediator are immobilized on the surface by the polymer. In certain instances, one or more of the nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, NAD(P)+-dependent dehydrogenase, NAD(P)H oxidoreductase and redox mediator are covalently bonded to the polymer. In certain embodiments, the NAD(P)+-dependent dehydrogenase is glucose dehydrogenase, alcohol dehydrogenase or D-3-hydroxybutyrate dehydrogenase. In some embodiments, the NAD(P)H oxidoreductase

is diaphorase. Also provided are analyte sensors having the subject enzyme compositions disposed proximate to a working electrode for monitoring an analyte level in a subject as well as methods for fabricating an analyte sensor and methods for using the analyte sensors in monitoring an analyte, such as glucose, an alcohol or 13-hydroxybutyrate in a subject.

BRIEF DESCRIPTION OF THE DRAWINGS

A detailed description of various embodiments of the present disclosure is provided herein with reference to the accompanying drawings, which are briefly described below. The drawings are illustrative and are not necessarily drawn to scale. The drawings illustrate various embodiments of the present disclosure and may illustrate one or more embodiment(s) or example(s) of the present disclosure in whole or in part. A reference numeral, letter, and/or symbol that is used in one drawing to refer to a particular element may be used in another drawing to refer to a like element.

FIG. 1 shows the signal output for a D-3-hydroxybutyrate dehydrogenase sensor over the course of 2.3 hours at varying concentrations of D-3-hydroxybutyrate according to certain embodiments.

FIG. 2 depicts the linearity of the sensor signal of a D-3-hydroxybutyrate dehydrogenase sensor as a function of D-3-hydroxybutyrate concentration.

FIG. 3 shows the signal output for a D-3-hydroxybutyrate dehydrogenase sensor using free NAD over the course of 3.6 hours at varying concentrations of D-3-hydroxybutyrate according to certain embodiments.

FIG. 4 depicts the linearity of the sensor signal of a D-3-hydroxybutyrate dehydrogenase sensor as a function of D-3-hydroxybutyrate concentration (Ketone).

FIG. 5 depicts the stability of the sensor signal of a D-3-hydroxybutyrate dehydrogenase sensor.

FIG. 6 depicts the stability of the sensor signal of a free NAD and immobilized NAD sensor.

DETAILED DESCRIPTION

Before the embodiments of the present disclosure are described, it is to be understood that this invention is not limited to particular embodiments described, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting, since the scope of the embodiments of the invention will be embodied by the appended claims.

Where a range of values is provided, it is understood that each intervening value, to the tenth of the unit of the lower limit unless the context clearly dictates otherwise, between the upper and lower limits of that range is also specifically disclosed. Each smaller range between any stated value or intervening value in a stated range and any other stated or intervening value in that stated range is encompassed within the invention. The upper and lower limits of these smaller ranges may independently be included or excluded in the range, and each range where either, neither or both limits are included in the smaller ranges is also encompassed within the invention, subject to any specifically excluded limit in the stated range. Where the stated range includes one or both of the limits, ranges excluding either or both of those included limits are also included in the invention.

In the description of the invention herein, it will be understood that a word appearing in the singular encompasses its plural counterpart, and a word appearing in the

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plural encompasses its singular counterpart, unless implicitly or explicitly understood or stated otherwise. Merely by way of example, reference to “an” or “the” “enzymes” encompasses a single enzyme, as well as a combination and/or mixture of two or more different enzymes, reference to “a” or “the” “concentration value” encompasses a single concentration value, as well as two or more concentration values, and the like, unless implicitly or explicitly understood or stated otherwise. Further, it will be understood that for any given component described herein, any of the possible candidates or alternatives listed for that component, may generally be used individually or in combination with one another, unless implicitly or explicitly understood or stated otherwise. Additionally, it will be understood that any list of such candidates or alternatives is merely illustrative, not limiting, unless implicitly or explicitly understood or stated otherwise.

Various terms are described below to facilitate an understanding of the invention. It will be understood that a corresponding description of these various terms applies to corresponding linguistic or grammatical variations or forms of these various terms. It will also be understood that the invention is not limited to the terminology used herein, or the descriptions thereof, for the description of particular embodiments. Merely by way of example, the invention is not limited to particular lactates, bodily or tissue fluids, blood or capillary blood, or sensor constructs or usages, unless implicitly or explicitly understood or stated otherwise, as such may vary.

The publications discussed herein are provided solely for their disclosure prior to the filing date of the application. Nothing herein is to be construed as an admission that the embodiments of the invention are not entitled to antedate such publication by virtue of prior invention. Further, the dates of publication provided may be different from the actual publication dates which may need to be independently confirmed.

The present disclosure discloses enzyme compositions that include nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof and an electron transfer agent having a transition metal complex. In some embodiments, the subject enzyme compositions include nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, an NAD(P)+-dependent dehydrogenase, an NAD(P)H oxidoreductase and an electron transfer agent having a transition metal complex, and analyte sensors with enzyme layers that include immobilized NAD(P)+ or derivative thereof and an electron transfer agent comprising a transition metal complex. Embodiments of the present disclosure relate to enzyme compositions for analyte sensing including where the subject compositions provide for monitoring of an analyte in vivo over an extended period of time. Where the subject enzyme compositions include an NAD(P)+-dependent dehydrogenase, analyte sensors described herein provide for clinically accurate electrochemical measurement of analytes that are catalyzed by an NAD(P)+-dependent dehydrogenase. As described in greater detail below, the subject enzyme compositions provide for clinically accurate electrochemical measurement of analytes as measured by Clark error grid analysis and/or MARD analysis and/or MAD analysis. In particular, the subject enzyme compositions provide for measurement by an analyte sensor incorporating the subject compositions to produce a signal that increases linearly as a function of analyte concentration.

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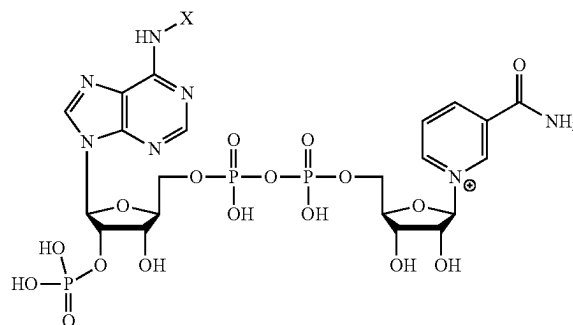
In addition, the subject enzyme compositions provide for clinically accurate electrochemical measurement of analytes that are catalyzed by an NAD(P)+-dependent dehydrogenase within 30 seconds of contacting a fluidic sample (e.g., interstitial fluid when the sensor is positioned beneath the surface of a subject's skin) with the sensor. In certain instances, the subject enzyme compositions provide for clinically accurate electrochemical measurements of analytes that are catalyzed by an NAD(P)+-dependent dehydrogenase immediately after contacting the fluidic sample with the sensor.

The subject enzyme compositions include an NAD(P)+-dependent dehydrogenase, such as glucose dehydrogenase, alcohol dehydrogenase or D-3-hydroxybutyrate dehydrogenase. In some embodiments, NAD(P)+-dependent dehydrogenases of interest are oxidoreductases belonging to the enzyme class 1.1.1.

The NAD(P)+-dependent dehydrogenase may be present in the subject compositions in an amount that varies, such as from 0.05 μg to 5 μg , such as from 0.1 μg to 4 μg , such as from 0.2 μg to 3 μg and including from 0.5 μg to 2 μg . As such, the amount of NAD(P)+-dependent dehydrogenase is from 0.01% to 10% by weight of the total enzyme composition, such as from 0.05% to 9.5% by weight, such as from 0.1% to 9% by weight, such as 0.5% to 8.5% by weight, such as from 1% to 8% by weight and including from 2% to 7% by weight of the total enzyme composition.

Enzyme compositions also include nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof. In some embodiments, enzyme compositions of interest include nicotinamide adenine dinucleotide phosphate (NAD(P)+). In other embodiments, enzyme compositions include a derivative of nicotinamide adenine dinucleotide phosphate (NAD(P)+). Derivatives of nicotinamide adenine dinucleotide phosphate (NAD(P)+) may include compounds of Formula I:

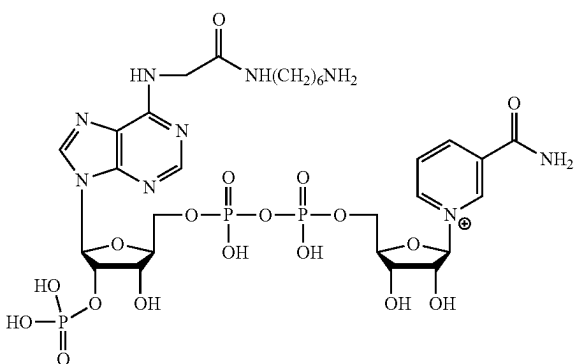
Formula I



where X is alkyl, substituted alkyl, aryl, substituted aryl, acyl, and aminoacyl.

In some embodiments, X is an aminoacyl substituted alkyl. In some embodiments, X is $\text{CH}_2\text{C}(\text{O})\text{NH}(\text{CH}_2)_y\text{NH}_2$ where y is an integer from 1 to 10, such as 2 to 9, such as 3 to 8 and including where y is 6. In certain instances, X is $\text{CH}_2\text{C}(\text{O})\text{NH}(\text{CH}_2)_6\text{NH}_2$. In these embodiments, the derivative of nicotinamide adenine dinucleotide phosphate (NAD(P)+) in the subject enzyme composition is:

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Embodiments of the enzyme composition also include an NAD(P)H oxidoreductase. In certain embodiments, the enzyme composition includes diaphorase. The amount of NAD(P)H oxidoreductase (e.g., diaphorase) present in the subject compositions ranges from 0.01 μ g to 10 μ g, such as from 0.02 μ g to 9 μ g, such as from 0.03 μ g to 8 μ g, such as from 0.04 μ g to 7 μ g, such as from 0.05 μ g to 5 μ g, such as from 0.1 μ g to 4 μ g, such as from 0.2 μ g to 3 μ g and including from 0.5 μ g to 2 μ g. As such, the amount of NAD(P)H oxidoreductase (e.g., diaphorase) is from 0.01% to 10% by weight of the total enzyme composition, such as from 0.05% to 9.5% by weight, such as from 0.1% to 9% by weight, such as 0.5% to 8.5% by weight, such as from 1% to 8% by weight and including from 2% to 7% by weight of the total enzyme composition.

In some embodiments, the weight ratio of NAD(P)+-dependent dehydrogenase to NAD(P)H oxidoreductase (e.g., diaphorase) ranges from 1 to 10 NAD(P)+-dependent dehydrogenase to NAD(P)H oxidoreductase, such as from 1 to 8, such as from 1 to 5, such as from 1 to 2 and including from 1 to 1 NAD(P)+-dependent dehydrogenase to NAD(P)H oxidoreductase. In other embodiments, the weight ratio of NAD(P)+-dependent dehydrogenase to NAD(P)H oxidoreductase ranges from 10 to 1 NAD(P)+-dependent dehydrogenase to NAD(P)H oxidoreductase, such as from 8 to 1, such as from 5 to 1 and including from 2 to 1 NAD(P)+-dependent dehydrogenase to NAD(P)H oxidoreductase.

Enzyme compositions of interest also include an electron transfer agent having an transition metal complex. They may be electroreducible and electrooxidizable ions or molecules having redox potentials that are a few hundred millivolts above or below the redox potential of the standard calomel electrode (SCE). Examples of transition metal complexes include metallocenes including ferrocene, hexacyanoferrate (II), ruthenium hexamine, etc. Additional examples include those described in U.S. Pat. Nos. 6,736,957, 7,501,053 and 7,754,093, the disclosures of each of which are incorporated herein by reference in their entirety.

In some embodiments, electron transfer agents are osmium transition metal complexes with one or more ligands, each ligand having a nitrogen-containing heterocycle such as 2,2'-bipyridine, 1,10-phenanthroline, 1-methyl, 2-pyridyl biimidazole, or derivatives thereof. The electron transfer agents may also have one or more ligands covalently bound in a polymer, each ligand having at least one nitrogen-containing heterocycle, such as pyridine, imidazole, or derivatives thereof. One example of an electron transfer agent includes (a) a polymer or copolymer having pyridine or imidazole functional groups and (b) osmium cations complexed with two ligands, each ligand containing 2,2'-bipyridine, 1,10-phenanthroline, or derivatives thereof.

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the two ligands not necessarily being the same. Some derivatives of 2,2'-bipyridine for complexation with the osmium cation include but are not limited to 4,4'-dimethyl-2,2'-bipyridine and mono-, di-, and polyalkoxy-2,2'-bipyridines, including 4,4'-dimethoxy-2,2'-bipyridine. Derivatives of 1,10-phenanthroline for complexation with the osmium cation include but are not limited to 4,7-dimethyl-1,10-phenanthroline and mono, di-, and polyalkoxy-1,10-phenanthrolines, such as 4,7-dimethoxy-1,10-phenanthroline. Polymers for complexation with the osmium cation include but are not limited to polymers and copolymers of poly(1-vinyl imidazole) (referred to as "PVI") and poly(4-vinyl pyridine) (referred to as "PVP"). Suitable copolymer substituents of poly(1-vinyl imidazole) include acrylonitrile, acrylamide, and substituted or quaternized N-vinyl imidazole, e.g., electron transfer agents with osmium complexed to a polymer or copolymer of poly(1-vinyl imidazole).

The subject enzyme compositions may be heterogeneous or homogenous. In some embodiments, each component (i.e., nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, an NAD(P)+-dependent dehydrogenase, an NAD(P)H oxidoreductase and an electron transfer agent having a transition metal complex) is uniformly distributed throughout the composition, e.g., when applied to an electrode, as described in greater detail below. For example, each of nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, an NAD(P)+-dependent dehydrogenase, an NAD(P)H oxidoreductase and an electron transfer agent having a transition metal complex may be distributed uniformly throughout the composition, such that the concentration of each component is the same throughout.

In certain embodiments, the subject enzyme compositions described herein are polymeric. Polymers that may be used may be branched or unbranched and may be homopolymers formed from the polymerization of a single type of monomer or heteropolymers that include two or more different types of monomers. Heteropolymers may be copolymers where the copolymer has alternating monomer subunits, or in some cases, may be block copolymers, which include two or more homopolymer subunits linked by covalent bonds (e.g., diblock or triblock copolymers). In some embodiments, the subject enzyme compositions include a heterocycle-containing polymer. The term heterocycle (also referred to as "heterocyclyl") is used herein in its conventional sense to refer to any cyclic moiety which includes one or more heteroatoms (i.e., atoms other than carbon) and may include, but are not limited to N, P, O, S, Si, etc. Heterocycle-containing polymers may be heteroalkyl, heteroalkanyl, heteroalkenyl and heteroalkynyl as well as heteroaryl or heteroaryalkyl.

"Heteroalkyl, Heteroalkanyl, Heteroalkenyl and Heteroalkynyl" by themselves or as part of another substituent refer to alkyl, alkanyl, alkenyl and alkynyl groups, respectively, in which one or more of the carbon atoms (and any associated hydrogen atoms) are independently replaced with the same or different heteroatomic groups. Typical heteroatomic groups which can be included in these groups include, but are not limited to, —O—, —S—, —N=N—, —N—N—, —P(O)—, —P(O)2—, —SO2—, —SnR4— and the like, where R is independently hydrogen, alkyl, substituted alkyl, aryl, substituted aryl, arylalkyl, substituted arylalkyl, cycloalkyl, substituted cycloalkyl, cycloheteroalkyl, substituted cycloheteroalkyl, heteroalkyl, substituted

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heteroalkyl, heteroaryl, substituted heteroaryl, heteroarylalkyl or substituted heteroarylalkyl.

"Heteroaryl" by itself or as part of another substituent, refers to a monovalent heteroaromatic radical derived by the removal of one hydrogen atom from a single atom of a heteroaromatic ring system. Typical heteroaryl groups include, but are not limited to, groups derived from acridine, arindole, carbazole, β -carboline, chromane, chromene, cinoline, furan, imidazole, indazole, indole, indoline, indolizine, isobenzofuran, isochromene, isoindole, isoindoline, isoquinoline, isothiazole, isoxazole, naphthyridine, oxadiazole, oxazole, perimidine, phenanthridine, phenanthroline, phenazine, phthalazine, pteridine, purine, pyran, pyrazine, pyrazole, pyridazine, pyridine, pyrimidine, pyrrole, pyrrolizine, quinazoline, quinoline, quinolizine, quinoxaline, tetrazole, thiadiazole, thiazole, thiophene, triazole, xanthene, benzodioxole and the like. In certain embodiments, the heteroaryl group is from 5-20 membered heteroaryl. In certain embodiments, the heteroaryl group is from 5-10 membered heteroaryl. In certain embodiments, heteroaryl groups are those derived from thiophene, pyrrole, benzothiophene, benzofuran, indole, pyridine, quinoline, imidazole, oxazole and pyrazine.

"Heteroarylalkyl" by itself or as part of another substituent, refers to an acyclic alkyl radical in which one of the hydrogen atoms bonded to a carbon atom, typically a terminal or sp^3 carbon atom, is replaced with a heteroaryl group. Where specific alkyl moieties are intended, the nomenclature heteroarylalkanyl, heteroarylalkenyl and/or heteroarylalkynyl is used. In certain embodiments, the heteroarylalkyl group is a 6-30 membered heteroarylalkyl, e.g., the alkanyl, alkenyl or alkynyl moiety of the heteroarylalkyl is 1-10 membered and the heteroaryl moiety is a 5-20-membered heteroaryl. In certain embodiments, the heteroarylalkyl group is 6-20 membered heteroarylalkyl, e.g., the alkanyl, alkenyl or alkynyl moiety of the heteroarylalkyl is 1-8 membered and the heteroaryl moiety is a 5-12-membered heteroaryl.

In some embodiments, the heterocycle-containing component is an aromatic ring system. "Aromatic Ring System" by itself or as part of another substituent, refers to an unsaturated cyclic or polycyclic ring system having a conjugated 7C electron system. Specifically included within the definition of "aromatic ring system" are fused ring systems in which one or more of the rings are aromatic and one or more of the rings are saturated or unsaturated, such as, for example, fluorene, indane, indene, phenalene, etc. Typical aromatic ring systems include, but are not limited to, acenaphthylene, acenaphthylene, acephenanthrylene, anthracene, azulene, benzene, chrysene, coronene, fluoranthene, fluorene, hexacene, hexaphene, hexalene, as-indacene, s-indacene, indane, indene, naphthalene, octacene, octaphene, octalene, ovalene, penta-2,4-diene, pentacene, pentalene, pentaphene, perylene, phenalene, phenanthrene, picene, pleiadene, pyrene, pyranthrene, rubicene, triphenylene, trinaphthalene and the like.

"Heteroaromatic Ring System" by itself or as part of another substituent, refers to an aromatic ring system in which one or more carbon atoms (and any associated hydrogen atoms) are independently replaced with the same or different heteroatom. Typical heteroatoms to replace the carbon atoms include, but are not limited to, N, P, O, S, Si, etc. Specifically included within the definition of "heteroaromatic ring systems" are fused ring systems in which one or more of the rings are aromatic and one or more of the rings are saturated or unsaturated, such as, for example, arindole, benzodioxan, benzofuran, chromane, chromene, indole,

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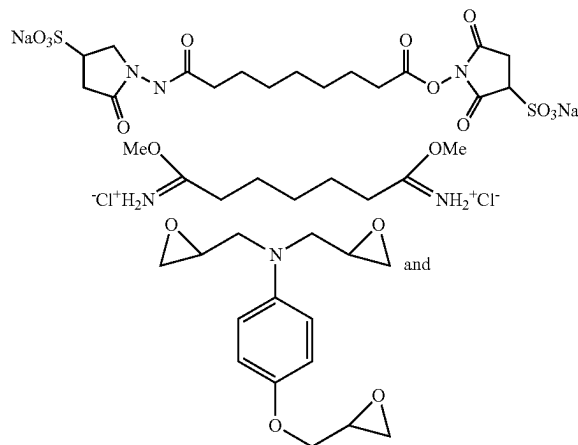
indoline, xanthene, etc. Typical heteroaromatic ring systems include, but are not limited to, arindole, carbazole, β -carboline, chromane, chromene, cinoline, furan, imidazole, indazole, indole, indoline, indolizine, isobenzofuran, isochromene, isoindole, isoindoline, isoquinoline, isothiazole, isoxazole, naphthyridine, oxadiazole, oxazole, perimidine, phenanthridine, phenanthroline, phenazine, phthalazine, pteridine, purine, pyran, pyrazine, pyrazole, pyridazine, pyridine, pyrimidine, pyrrole, pyrrolizine, quinazoline, quinoline, quinolizine, quinoxaline, tetrazole, thiadiazole, thiazole, thiophene, triazole, xanthene and the like.

In certain embodiments, enzyme compositions of interest include a heterocyclic nitrogen containing component, such as polymers of polyvinylpyridine (PVP) and polyvinylimidazole.

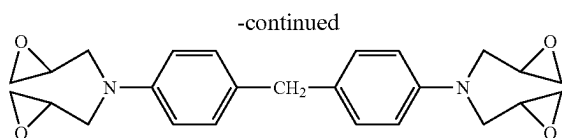
The polymeric enzyme compositions may also include one or more crosslinkers (crosslinking agent) such that the polymeric backbone enzyme composition is crosslinked. As described herein, reference to linking two or more different polymers together is intermolecular crosslinking, whereas linking two more portions of the same polymer is intramolecular crosslinking. In embodiments of the present disclosure, crosslinkers may be capable of both intermolecular and intramolecular crosslinkings at the same time.

Suitable crosslinkers may be bifunctional, trifunctional or tetrafunctional, each having straight chain or branched structures. Crosslinkers having branched structures include a multi-arm branching component, such as a 3-arm branching component, a 4-arm branching component, a 5-arm branching component, a 6-arm branching component or a larger number arm branching component, such as having 7 arms or more, such as 8 arms or more, such as 9 arms or more, such as 10 arms or more and including 15 arms or more. In certain instances, the multi-arm branching component is a multi-arm epoxide, such as 3-arm epoxide or a 4-arm epoxide. Where the multi-arm branching component is a multi-arm epoxide, the multi-arm branching component may be a polyethylene glycol (PEG) multi-arm epoxide or a non-polyethylene glycol (non-PEG) multi-arm epoxide. In some embodiments, the multi-arm branching component is a non-PEG multi-arm epoxide. In other embodiments, the multi-arm branching component is a PEG multi-arm epoxide. In certain embodiments, the multi-arm branching component is a 3-arm PEG epoxide or a 4-arm PEG epoxide.

Examples of crosslinkers include but are not limited to polyethylene glycol diglycidyl ether, N,N-diglycidyl-4-glycidylloxylaniline as well as nitrogen-containing multi-functional crosslinkers having the structures:



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In some instances, one or more bonds with the one or more components of the enzyme composition may be formed such as between one or more of the nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, NAD(P)+-dependent dehydrogenase, NAD(P)H oxidoreductase and electron transfer agent. By bonds is meant any type of an interaction between atoms or molecules that allows chemical compounds to form associations with each other, such as, but not limited to, covalent bonds, ionic bonds, dipole-dipole interactions, hydrogen bonds, London dispersion forces, and the like. For example, in situ polymerization of the enzyme compositions can form cross-links between the polymers of the composition and the NAD(P)+-dependent dehydrogenase, nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, the NAD(P)H oxidoreductase and the electron transfer agent. In certain embodiments, crosslinking of the polymer to the one or more of the NAD(P)+-dependent dehydrogenase, nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, the NAD(P)H oxidoreductase and the electron transfer agent facilitates a reduction in the occurrence of delamination of the enzyme compositions from an electrode.

As described herein, the subject enzyme may be used in an analyte sensor to monitor the concentration of an NAD(P)+-dependent dehydrogenase analyte, such as glucose, an alcohol, a ketone, lactate, or β -hydroxybutyrate, and the sensor may have one or more electrodes with the enzyme composition. In embodiments, the analyte sensor includes: a working electrode having a conductive material the subject enzyme composition proximate to (e.g., disposed on) and in contact with the conductive material. One or more other electrode may be included such as one or more counter electrodes, one or more reference electrodes and/or one or more counter/reference electrodes.

The particular configuration of electrochemical sensors may depend on the use for which the analyte sensor is intended and the conditions under which the analyte sensor will operate. In certain embodiments of the present disclosure, analyte sensors are in vivo wholly positioned analyte sensors or transcutaneously positioned analyte sensors configured for in vivo positioning in a subject. In one example, at least a portion of the sensor may be positioned in the subcutaneous tissue for testing lactate concentrations in interstitial fluid. In another example, at least a portion of the sensor may be positioned in the dermal tissue for testing analyte concentration in dermal fluid.

In embodiments, one or more of the subject enzyme compositions is positioned proximate to (e.g., disposed on) the surface of a working electrode. In some instances, a plurality of enzyme compositions are positioned proximate to the surface of working electrode (e.g., in the form of spots). In certain cases, a discontinuous or continuous perimeter is formed around each of the plurality of enzyme compositions positioned proximate to the surface of the working electrode. Examples of depositing a plurality of reagent compositions to the surface of an electrode as well as forming a discontinuous or continuous perimeter around each reagent composition is described in U.S. Patent Pub-

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lication No. 2012/0150005 and in U.S. Patent Application No. 62/067,813, the disclosures of which are herein incorporated by reference.

The subject enzyme compositions having nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, NAD(P)+-dependent dehydrogenase, NAD(P)H oxidoreductase and electron transfer agent may be deposited onto the surface of the working electrode as one large application which covers the desired portion of the working electrode or in the form of an array of a plurality of enzyme compositions, e.g., spaced apart from each other. Depending upon use, any or all of the enzyme compositions in the array may be the same or different from one another. For example, an array may include two or more, 5 or more enzyme composition array features containing nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, NAD(P)+-dependent dehydrogenase, NAD(P)H oxidoreductase and electron transfer agent, 10 or more, 25 or more, 50 or more, 100 or more, or even 1000 or more, in an area of 100 mm² or less, such as 75 mm² or less, or 50 mm² or less, for instance 25 mm² or less, or 10 mm² or less, or 5 mm² or less, such as 2 mm² or less, or 1 mm² or less, 0.5 mm² or less, or 0.1 mm² or less.

The shape of deposited enzyme composition may vary within or between sensors. For example, in certain embodiments, the deposited membrane is circular. In other embodiments, the shape will be of a triangle, square, rectangle, circle, ellipse, or other regular or irregular polygonal shape (e.g., when viewed from above) as well as other two-dimensional shapes such as a circle, half circle or crescent shape. All or a portion of the electrode may be covered by the enzyme composition, such as 5% or more, such as 25% or more, such as 50% or more, such as 75% or more and including 90% or more. In certain instances, the entire electrode surface is covered by the enzyme composition (i.e., 100%).

Fabricating an electrode and/or sensor according to embodiments of the present disclosure produces a reproducible enzyme composition deposited on the surface of the electrode. For example, enzyme compositions provided herein may deviate from each other by 5% or less, such as by 4% or less, such as by 3% or less, such as by 2% or less, such as by 1% or less and including by 0.5% or less. In some embodiments, the sensing composition includes nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof and an electron transfer agent. In certain embodiments, deposited enzyme compositions containing nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, NAD(P)+-dependent dehydrogenase, NAD(P)H oxidoreductase and electron transfer agent show no deviation from one another and are identical.

In certain embodiments, methods further include drying the enzyme composition deposited on the electrode. Drying may be performed at room temperature, at an elevated temperature, as desired, such as at a temperature ranging from 25° C. to 100° C., such as from 30° C. to 80° C. and including from 40° C. to 60° C.

Examples of configurations for the subject analyte sensors and methods for fabricating them may include, but are not limited to, those described in U.S. Pat. Nos. 6,175,752, 6,134,461, 6,579,690, 6,605,200, 6,605,201, 6,654,625, 6,746,582, 6,932,894, 7,090,756, 5,356,786, 6,560,471, 5,262,035, 6,881,551, 6,121,009, 6,071,391, 6,377,894, 6,600,997, 6,514,460, 5,820,551, 6,736,957, 6,503,381, 6,676,816, 6,514,718, 5,593,852, 6,284,478, 7,299,082, 7,811,231, 7,822,557, 8,106,780, and 8,435,682; U.S. Patent Application Publication Nos. 2010/0198034, 2010/0324392,

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2010/0326842, 2007/0095661, 2010/0213057, 2011/0120865, 2011/0124994, 2011/0124993, 2010/0213057, 2011/0213225, 2011/0126188, 2011/0256024, 2011/0257495, 2012/0157801, 2012/0245447, 2012/0157801, 2012/0323098, and 20130116524, the disclosures of each of which are incorporated herein by reference in their entirety.

In some embodiments, in vivo sensors may include an insertion tip positionable below the surface of the skin, e.g., penetrating through the skin and into, e.g., the subcutaneous space, in contact with the user's biological fluid such as interstitial fluid. Contact portions of working electrode, a reference electrode and a counter electrode are positioned on the first portion of the sensor situated above the skin surface. A working electrode, a reference electrode and a counter electrode are positioned at the inserted portion of the sensor. Traces may be provided from the electrodes at the tip to a contact configured for connection with sensor electronics.

In certain embodiments, the working electrode and counter electrode of the sensor as well as dielectric material of are layered. For example, the sensor may include a non-conductive material layer, and a first conductive layer such as conductive polymer, carbon, platinum-carbon, gold, etc., disposed on at least a portion of the non-conductive material layer (as described above). The enzyme composition is positioned on one or more surfaces of the working electrode, or may otherwise be directly or indirectly contacted to the working electrode. A first insulation layer, such as a first dielectric layer may be disposed or layered on at least a portion of a first conductive layer and a second conductive layer may be positioned or stacked on top of at least a portion of a first insulation layer (or dielectric layer). The second conductive layer may be a reference electrode. A second insulation layer, such as a second dielectric layer may be positioned or layered on at least a portion of the second conductive layer. Further, a third conductive layer may be positioned on at least a portion of the second insulation layer and may be a counter electrode. Finally, a third insulation layer may be disposed or layered on at least a portion of the third conductive layer. In this manner, the sensor may be layered such that at least a portion of each of the conductive layers is separated by a respective insulation layer (for example, a dielectric layer).

In other embodiments, some or all of the electrodes may be provided in a co-planar manner such that two or more electrodes may be positioned on the same plane (e.g., side-by side (e.g., parallel) or angled relative to each other) on the material. For example, co-planar electrodes may include a suitable spacing there between and/or include a dielectric material or insulation material disposed between the conductive layers/electrodes. Furthermore, in certain embodiments one or more of the electrodes may be disposed on opposing sides of the non-conductive material. In such embodiments, electrical contact may be on the same or different sides of the non-conductive material. For example, an electrode may be on a first side and its respective contact may be on a second side, e.g., a trace connecting the electrode and the contact may traverse through the material. A via provides an avenue through which an electrical trace is brought to an opposing side of a sensor.

The subject analyte sensors be configured for monitoring the level of an analyte (e.g., glucose, an alcohol, a ketone, lactate, β -hydroxybutyrate) over a time period which may range from seconds, minutes, hours, days, weeks, to months, or longer.

In certain embodiments, the analyte sensor includes a mass transport limiting layer, e.g., an analyte flux modulating layer, to act as a diffusion-limiting barrier to reduce the

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rate of mass transport of the analyte, for example, glucose, an alcohol, a ketone, lactate, β -hydroxybutyrate, when the sensor is in use. The mass transport limiting layers limit the flux of an analyte to the electrode in an electrochemical sensor so that the sensor is linearly responsive over a large range of analyte concentrations. Mass transport limiting layers may include polymers and may be biocompatible. A mass transport limiting layer may provide many functions, e.g., biocompatibility and/or interferent-eliminating functions, etc., or functions may be provided by various membrane layers.

In certain embodiments, a mass transport limiting layer is a membrane composed of crosslinked polymers containing heterocyclic nitrogen groups, such as polymers of polyvinylpyridine and polyvinylimidazole. Embodiments also include membranes that are made of a polyurethane, or polyether urethane, or chemically related material, or membranes that are made of silicone, and the like.

The membrane may be formed by crosslinking in situ a polymer, modified with a zwitterionic moiety, a non-pyridine copolymer component, and optionally another moiety that is either hydrophilic or hydrophobic, and/or has other desirable properties, in an alcohol-buffer solution. The modified polymer may be made from a precursor polymer containing heterocyclic nitrogen groups. For example, a precursor polymer may be polyvinylpyridine or polyvinylimidazole. Optionally, hydrophilic or hydrophobic modifiers may be used to "fine-tune" the permeability of the resulting membrane to an analyte of interest. Optional hydrophilic modifiers, such as poly(ethylene glycol), hydroxyl or polyhydroxyl modifiers, may be used to enhance the biocompatibility of the polymer or the resulting membrane.

Suitable mass transport limiting membranes in the subject analyte sensors may include, but are not limited to those described in U.S. Pat. No. 6,932,894, the disclosure of which is herein incorporated by reference. In certain embodiments, the mass transport limiting membrane is a SMART membrane that is temperature independent. Suitable temperature independent membranes may include, but are not limited to those described in U.S. Patent Publication No. 2012/0296186 and U.S. patent application Ser. No. 14/737,082, the disclosures of which are herein incorporated by reference.

Analyte sensors according to certain embodiments may be configured to operate at low oxygen concentration. By low oxygen concentration is meant the concentration of oxygen is 1.5 mg/L or less, such as 1.0 mg/L or less, such as 0.75 mg/L or less, such as 0.6 mg/L or less, such as 0.3 mg/L or less, such as 0.25 mg/L or less, such as 0.15 mg/L or less, such as 0.1 mg/L or less and including 0.05 mg/L or less.

Aspects of the present disclosure also include methods for in vivo monitoring analyte levels over time with analyte sensors incorporating an enzyme composition containing nicotinamide adenine dinucleotide phosphate (NAD(P)+) or derivative thereof, NAD(P)+-dependent dehydrogenase, NAD(P)H oxidoreductase and electron transfer agent. Generally, in vivo monitoring the concentration of analyte in a fluid of the body of a subject includes inserting at least partially under a skin surface an in vivo analyte sensor as disclosed herein, contacting the monitored fluid (interstitial, blood, dermal, and the like) with the inserted sensor, and

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generating a sensor signal at the working electrode. The presence and/or concentration of analyte detected by the analyte sensor may be displayed, stored, forwarded, and/or otherwise processed. A variety of approaches may be employed to determine the concentration of analyte (e.g., glucose, an alcohol, a ketone, lactate, β -hydroxybutyrate) with the subject sensors. In certain aspects, an electrochemical analyte concentration monitoring approach is used. For example, monitoring the concentration of analyte using the sensor signal may be performed by coulometric, amperometric, voltammetric, potentiometric, or any other convenient electrochemical detection technique.

These methods may also be used in connection with a device that is used to detect and/or measure another analyte, including glucose, oxygen, carbon dioxide, electrolytes, or other moieties of interest, for example, or any combination thereof, found in a bodily fluid, including subcutaneous e.g., interstitial fluid, dermal fluid, blood or other bodily fluid of interest or any combination thereof.

In certain embodiments, the method further includes attaching an electronics unit to the skin of the patient, coupling conductive contacts of the electronics unit to contacts of the sensor, collecting data using the electronics unit regarding a level of analyte from signals generated by the sensor, and forwarding the collected data from electronics unit to a receiver unit, e.g., by RF. The receiver unit may be a mobile telephone. The mobile telephone may include an application related to the monitored analyte. In certain embodiments, analyte information is forwarded by RFID protocol, Bluetooth, and the like.

The analyte sensor may be positionable in a user for automatic analyte sensing, either continuously or periodically. Embodiments may include monitoring the level of the analyte over a time period which may range from seconds, minutes, hours, days, weeks, to months, or longer. Future analyte levels may be predicted based on information obtained, e.g., the current lactate level at time zero as well as an analyte rate of change.

The sensor electronics unit may automatically forward data from the sensor/electronics unit to one or more receiver units. The sensor data may be communicated automatically and periodically, such as at a certain frequency as data is obtained or after a certain time period of sensor data stored in memory. For example, sensor electronics coupled to an in vivo positioned sensor may collect the sensor data for a predetermined period of time and transmit the collected data periodically (e.g., every minute, five minutes, or other predetermined period) to a monitoring device that is positioned in range from the sensor electronics.

In other embodiments, the sensor electronics coupled to the in vivo positioned sensor may communicate with the receiving device non automatically manner and not set to any specific schedule. For example, the sensor data may be communicated from the sensor electronics to the receiving device using RFID technology, and communicated whenever the sensor electronics are brought into communication range of the analyte monitoring device. For example, the in vivo positioned sensor may collect sensor data in memory until the monitoring device (e.g., receiver unit) is brought into communication range of the sensor electronics unit—e.g., by the patient or user. When the in vivo positioned

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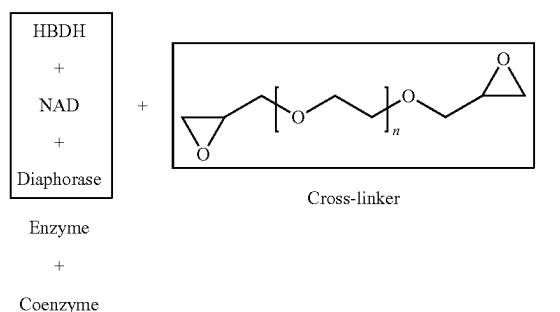
sensor is detected by the monitoring device, the device establishes communication with the analyte sensor electronics and uploads the sensor data that has been collected since the last transfer of sensor data, for instance. In this way, the patient does not have to maintain close proximity to the receiving device at all times, and instead, can upload sensor data when desired by bringing the receiving device into range of the analyte sensor. In yet other embodiments, a combination of automatic and non-automatic transfers of sensor data may be implemented in certain embodiments. For example, transfers of sensor data may be initiated when brought into communication range, and then continued on an automatic basis if continued to remain in communication range.

EXPERIMENTAL

The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how to make and use the embodiments of the invention, and are not intended to limit the scope of what the inventors regard as their invention nor are they intended to represent that the experiments below are all or the only experiments performed. Efforts have been made to ensure accuracy with respect to numbers used (e.g., amounts, temperature, etc.) but some experimental errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, molecular weight is weight average molecular weight, temperature is in degrees Centigrade, and pressure is at or near atmospheric.

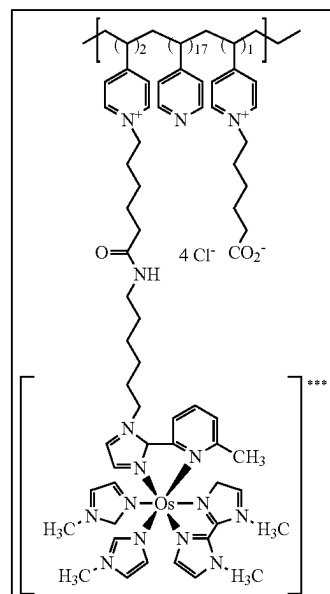
Example 1

Experiments were performed to demonstrate the performance of analyte sensors having a working electrode that contains nicotinamide adenine dinucleotide phosphate (NAD(P)⁺) or derivative thereof, NAD(P)⁺-dependent dehydrogenase, NAD(P)H oxidoreductase and electron transfer agent. The sensors were prepared by depositing onto the surface of an electrode an enzyme composition containing nicotinamide adenine dinucleotide phosphate, D-3-hydroxybutyrate dehydrogenase, diaphorase and a polymer bound osmium-transition metal catalyst and a difunctional cross-linker, as shown by the scheme (also referred to as a polymeric redox mediator in further examples):



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-continued



The sensors were tested in phosphate buffer containing varying concentrations of D-3-hydroxybutyrate. Table 1 summarizes beaker calibration and linearity of data signal from the prepared sensors.

TABLE 1

D-3-Hydroxybutyrate Sensor	
Slope	0.0163
R ²	0.9982

FIG. 1 shows the signal output over the course of 2.3 hours at varying concentrations of D-3-hydroxybutyrate (80 μ M, 160 μ M and 240 μ M) FIG. 2 depicts the linearity of the sensor signal as a function of D-3-hydroxybutyrate concentration. As shown in FIGS. 1 and 2, the sensor gives a linear and persistent response to D-3-hydroxybutyrate.

Example 2

Experiments were performed to demonstrate the performance of analyte sensors having a working electrode that contains free NAD. The sensor was prepared by depositing onto the surface of an electrode an enzyme composition containing the free NAD. Sensing layer formulation is described in Table 2. The sensing layer solutions was deposit on carbon electrodes, and cured at 25 C/60 H overnight, prior to addition of membrane. The membrane formulation is provided in Table 3. The sensor was dipped from above solution 3 $\times$ 5 mm/sec, and the sensor was cured at 25 C/60 H overnight, 56 C for two days.

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TABLE 2

Formulation Table Sensing Layer Solution					
	Vendor	Cat	mg/mL	Mixing uL	Final mg/mL
5 BD161212					
10 mM Hepes					
D-3-Hydroxybutyrate Dehydrogenase (HBD)	Toyobo	HBD-301	40	20	8.89
Diaphorase	Toyobo	DAD-311	40	20	8.89
10 NAD	Sigma	N0632	20	10	2.22
Polymeric redox mediator			40	20	8.89
Peg400			40	20	8.89
15 Total				90	

TABLE 3

Membrane Coating Membrane Solution					
Solution A	Vendor	Cat	MW	Mixing	
Poly(4-vinylpyridine) Ethanol/Hepes (10 mM pH 8.0)	Sigma	472352	106K	120 1	mg mL
Solution B	Vendor	Cat		Mixing	
Poly(ethylene glycol) diglycidyl ether (PEGDGE 400)	Poly- sciences	8210		100	mg
Ethanol/Hepes (10 mM pH 8.0)				1	mL
Final membrane solution				Mixing (mL)	
Solution A			4		
Solution B			0.4		

FIG. 3 shows the signal output over the course of 3.6 hours at varying concentrations of D-3-hydroxybutyrate (ketone). FIG. 4 depicts the linearity of the sensor signal as a function of D-3-hydroxybutyrate concentration. FIG. 5 shows sensor calibration at 10 mM. As shown in FIGS. 3-5, the sensor gives a linear and persistent response to D-3-hydroxybutyrate (ketone). Table 4 also shows the free NAD Ketone Sensor Beaker Calibration and Stability Summary.

TABLE 4

Slope	2.55
R ²	0.9996
Decay in 45 hours	-8%

Example 3

Experiments were also performed to demonstrate the performance of ketone sensors having a working electrode that contains free NAD versus immobilized NAD. The sensors were prepared by depositing onto the surface of an electrode an enzyme composition containing the free NAD (A) or the immobilized NAD (B). Sensing layer formulation is described in Table 5. The sensing layer solutions was deposit on carbon electrodes, and cured at 25 C/60 H overnight, prior to addition of membrane. The membrane formulation is provided in Table 6. The sensor was dipped (Table 7) from above solution 3 $\times$ 5 mm/sec, and the sensor

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was cured at 25 C/60 H overnight, 56 C for two days. FIG. 6 shows that both the free and immobilized NAD versions of the ketone sensor show similar stabilities and signals.

TABLE 5

Sensing Layer Solution							
BD161116	Vendor	Cat	mg/mL	Mixing uL		Final Mg/mL	
				A	B	A	B
10 mM Hepes							
D-3-Hydroxybutyrate Dehydrogenase (HBD)	Toyobo	HBD-301	40	20	20	8.89	8.89
Diaphorase	Toyobo	DAD-311	40	20	20	8.89	8.89
NAD	Sigma	NO632	20	10		2.22	0.00
NAD-NH ₂	Biotium		40		10	0.00	4.44
Glutaraldehyde			10	0	5	0.00	0.56
Polymeric redox mediator			40	20	20	8.89	8.89
Peg400			40	20	20	8.89	8.89
Total				90	95		

TABLE 6

Membrane Coating Membrane Solution				
Solution A	Vendor	MW	Mixing	
Smart Membrane Ethanol/Hepes (10 mM pH 8.0)	ADC	160K	120	mg mL
			1	
Solution B	Vendor	Cat	Mixing	
Poly(ethylene glycol) diglycidyl ether (PEGDGE 400)	Polysciences	8210	100	mg
Ethanol/Hepes (10 mM pH 8.0)			1	mL
Final membrane solution			Mixing (mL)	
Solution A			4	
Solution B			0.35	

TABLE 7

Membrane Dipping Condition (Free NAD sensor was coated thicker membrane than Immobilized NAD sensor)	
SL Solution	Dipping
A (Free NAD)	4 × 5
B (Immobilized NAD)	3 × 5

What is claimed is:

1. An electrochemical analyte sensor comprising:
 - a first portion configured to be positioned above a user's skin; and
 - a second portion configured to be transcutaneously positioned in the user's skin and in contact with the user's biological fluid to monitor the level of an analyte in vivo, the second portion comprising:

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- i) an electrode;
 - ii) an enzyme composition disposed upon the electrode, the enzyme composition comprising:
 - a) nicotinamide adenine dinucleotide (phosphate) (NAD(P)⁺) or derivative thereof;
 - b) D-3-hydroxybutyrate dehydrogenase;
 - c) an NAD(P)H oxidoreductase;
 - d) a polymeric redox mediator, and
 - iii) a membrane disposed on the enzyme composition, wherein the sensor produces a signal that increases linearly as a function of D-3-hydroxybutyrate concentration from 0 mM to 10 mM.
2. The analyte sensor of claim 1, wherein the signal increases linearly as a function of D-3-hydroxybutyrate concentration from 0 μM to 240 μM.
 3. The analyte sensor of claim 2, wherein the signal that increases linearly has a slope of about 0.016 nA/μM.
 4. The analyte sensor of claim 1, wherein the signal that increases linearly has a slope of about 2.5 nA/mM.
 5. The analyte sensor of claim 1, wherein the signal that increases linearly has a R² value of at least 0.99.
 6. The analyte sensor of claim 1, wherein the signal decays no more than 8% when exposed to a given D-3-hydroxybutyrate concentration for 45 hours.
 7. The analyte sensor of claim 1, wherein the NAD(P)H oxidoreductase is diaphorase.
 8. An electrochemical analyte sensor comprising:
 - a first portion configured to be positioned above a user's skin; and
 - a second portion configured to be transcutaneously positioned in the user's skin and in contact with the user's biological fluid to monitor the level of an analyte in vivo, the second portion comprising:
 - i) an electrode;
 - ii) an enzyme composition disposed upon the electrode, the enzyme composition comprising:
 - a) nicotinamide adenine dinucleotide (phosphate) (NAD(P)⁺) or derivative thereof;
 - b) D-3-hydroxybutyrate dehydrogenase;
 - c) an NAD(P)H oxidoreductase;
 - d) a polymeric redox mediator, and
 - iii) a membrane disposed on the enzyme composition, wherein the sensor produces a signal based on analyte concentration that decays no more than 8% when exposed to a given D-3-hydroxybutyrate concentration for 45 hours.
 9. The analyte sensor of claim 8, wherein the signal increases linearly as a function of D-3-hydroxybutyrate concentration.
 10. The analyte sensor of claim 9, wherein the signal increases linearly as a function of D-3-hydroxybutyrate concentration from 0 mM to 10 mM.
 11. The analyte sensor of claim 9, wherein the signal increases linearly as a function of D-3-hydroxybutyrate concentration from 0 μM to 240 μM.
 12. The analyte sensor of claim 11, wherein the signal that increases linearly has a slope of about 0.016 nA/μM.
 13. The analyte sensor of claim 10, wherein the signal that increases linearly has a slope of about 2.5 nA/mM.
 14. The analyte sensor of claim 9, wherein the signal that increases linearly has a R² value of at least 0.99.
 15. The analyte sensor of claim 8, wherein the NAD(P)H oxidoreductase is diaphorase.

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